The Blind Watchmaker

Harmonic Latent Space and the Geometry of Bias and Variance

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Abstract

Do you exist in other people's realities the same way you exist in your own? It is a question the field tends to defer: the reflex is to hand the burden of proof back to whoever raised it, as if demanding evidence were the neutral posture and the asking were the indulgence. That reflex is not neutral. It is a prior, learned so early it passes for common sense, and priors are precisely the subject of this paper. The dominant program in machine learning rests on one of them: that a single architecture, given enough parameters, data, and compute, can reach anything worth learning - that the architecture itself is neutral ground. We argue instead that what a model can learn is set less by how large it is than by the geometry of its representation - the shape of the space its weights are allowed to occupy. We make that concrete with two components of the Praxis framework [4]: a harmonic head that writes a learned standing wave over its features, and a byte-latent encoder [20], which finds that rhythm in content-adaptive byte patches. The same geometry reaches its finest, per-feature form in a plain transformer, so across these approaches it is the granularity that changes, not the geometry. In both, the classical bias-variance trade-off stops being a single dial. A static, population-average rhythm (the spectrum the field phase-locks to) is pure bias; an input-conditional modulation of that rhythm is structured variance; and because the two live on orthogonal axes of a harmonic latent space, they can be lowered together rather than traded against each other. This reframing motivates a central conjecture: consensus modeling - learning the mean of a corpus - scales linearly in the patterns it must express, while harmonic modeling, which composes patterns by interference, scales logarithmically, because K frequency components address exponentially many configurations. The framework exists so that the reader can test it.

1 Introduction

The dominant recipe for capability is scale. Fix a transformer [30], then grow it: more parameters, more tokens, more compute, with returns described by smooth scaling laws [15]. The recipe works, but it assumes the architecture itself imposes no ceiling on which patterns gradient descent can find - that everything worth learning is reachable from any sufficiently large model. This paper questions that assumption and offers a concrete alternative: that the *geometry* a model is allowed to represent, not its size, determines the patterns accessible to it during finite training.

To make the claim precise we need a vocabulary for "geometry," and the cleanest one available is the oldest tradeoff in statistics. A model that minimizes error by collapsing toward the average of its training data has high bias and low variance; a model that chases the particulars of each example has low bias and high variance. Classically these trade off along a single line: you choose a point on it. Our first contribution is to show that in a *continuous, harmonic* latent space this is no longer true. Bias and variance become separate coordinates, and a model can move in one without paying in the other.

We ground this in mechanisms that already run in Praxis, not in metaphor. The harmonic head (Section 2) builds a two-dimensional standing wave over (position, feature) from a learned grid of frequency amplitudes with irrational, frozen phases. A *static* amplitude grid produces the same field for every input: it is the corpus-average rhythm, maximum bias, a flat-by-construction prior. Making those amplitudes depend on the input adds *structured* variance - deviation that still respects the learned harmonic constraints. The CALM codec (Section 3) shows the same physics from the other side: a continuous-latent autoregressor that reconstructs sequences near-perfectly by phase-locking its latent features to a dominant rhythm. From these two we draw a scaling conjecture (Section 4): the mean costs linearly, interference costs logarithmically.

This paper therefore contributes: (1) a reframing of bias and variance as orthogonal axes of a harmonic latent space, grounded in a running implementation and its measured diagnostics; (2) a conjecture that consensus modeling scales linearly while harmonic modeling scales logarithmically, stated honestly as a conjecture; (3) the Praxis framework for testing it; and (4) a candid account of what is measured, what is conjectured, and what remains philosophy.

2 The Mean and the Manifold

2.1 The tradeoff that was supposed to be a line

The bias-variance decomposition treats generalization error as the sum of a term that grows when the model is too rigid to fit the signal (bias) and a term that grows when it is too flexible and fits the noise (variance). The received wisdom is that you cannot reduce both at once with a fixed budget: regularize harder and bias rises; add capacity and variance rises. The tradeoff is drawn as a U-shaped curve with a single minimum, and model selection is the act of finding the bottom of that U.

That picture is correct when the model has one knob mediating both terms - one scalar of effective complexity. It stops being correct when bias and variance are produced by *different parts* of the architecture that can be optimized independently. A harmonic latent space is exactly such an architecture.

2.2 The harmonic field: a measured instance

The harmonic head (Praxis, `heads/harmonic.py`) modulates a model's hidden states multiplicatively by a learned field b:

$$h \;\mapsto\; h\,\big(1+b\big), \qquad b[t,d] \;=\; \mathrm{IRFFT2}\Big(a[f_t,f_d]\,e^{\,i\varphi[f_t,f_d]}\,/\,f^{\alpha}\Big),$$

where t indexes position, d indexes feature, and $a[f_t, f_d]$ is a learnable grid of amplitudes over a two-dimensional frequency lattice. The phases φ are *not* learned. They are seeded by Weyl's equidistribution theorem [31]: cell (f_t, f_d) receives phase $2\pi\,\mathrm{frac}(f_t\pi + f_d e)$, which is equidistributed on the torus because $\{1, \pi, e\}$ are linearly independent over $\mathbb{Q}$. A radial $1/f^{\alpha}$ envelope imposes a pink-noise prior. Multiplicative coupling is deliberate: the upstream network cannot cancel the field by emitting $h(x) - b$, because the field scales features rather than adding to them, so learning is forced to flow *through* the field rather than around it.

Now read this through the bias-variance lens.

Bias is the static spectrum. With a fixed amplitude grid, $b[t, d]$ is identical for every input. It is a single rhythm imposed on all of the corpus at once - the population average. In the limit where the grid concentrates onto one cell, every feature oscillates to the same beat and the spectrum heatmap is flat: maximum bias, a model that has collapsed to the mean rhythm of its data. The head exposes this directly. `concentration()` reports Hoyer sparsity of the grid (1 = a single cell, 0 = uniform), and a live spectrum snapshot shows where the energy sits. Early in training the heatmap is indistinguishable from noise; under a smoothness prior it develops a clean low-frequency band - the corpus rhythm crystallizing as a measurable object, not a metaphor.

Variance is the input-conditional deviation. The decisive architectural step was to stop forcing one static field on a corpus that mixes registers - dialogue, code, prose. A static grid must compromise across all of them, and the compromise degenerates into a content-blind side channel that the rest of the model has to route around. The fix makes the amplitudes a baseline plus an input-conditional delta:

$$a \;=\; a_{\mathrm{static}} \;+\; \Delta_{\phi}(\mathrm{context}),$$

where Δ_ϕ is a small network reading multi-scale pooled context, in the spirit of delay embeddings [26], with its output zero-initialized so the field begins as the trained static prior and *anneals* into input dependence. The static baseline carries the bias; the delta carries the variance; an L_2 penalty on the delta keeps the baseline load-bearing. A diagnostic, `harmonic_delta_norm` = $\mathrm{rms}(\Delta)/\mathrm{rms}(a_{\mathrm{static}})$, reads how much variance the model is choosing to add.

Zero means pure bias; a small bounded value means structured variance on top of a stable prior.

The point is that these are *separate parameters with separate gradients*. The smoothness prior controls the bias term; the delta network and its regularizer control the variance term. Lowering one does not mechanically raise the other. The U-shaped curve has become a two-dimensional manifold, and we have coordinates on it.

2.3 Temperature as a coordinate, not as noise

Sampling temperature is usually described as noise injection. On the bias-variance manifold it is better described as a position dial. At low temperature the model commits to the learned rhythm - the high-bias corner. As temperature rises it explores within the basin the harmonic constraints define - structured variance. Past a critical point it begins to violate those constraints, which reads as either failure or discovery depending on what you were hoping for. CALM makes this literal: rather than perturbing a latent continuously, it votes over decoded patches [22], so temperature selects how many candidate latents must agree before a patch is emitted. Temperature is not adding randomness to a fixed distribution; it is choosing where on the manifold to stand.

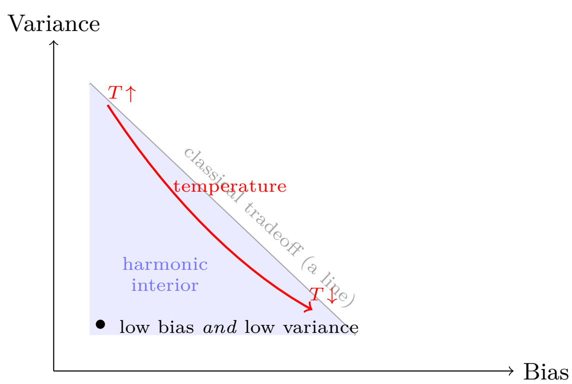

Figure 1: Bias and variance as orthogonal coordinates. Classically a model must sit somewhere on the gray line: less of one means more of the other. When a static spectrum supplies bias and an input-conditional modulation supplies variance, the two decouple, and the interior (shaded) becomes reachable - lower bias *and* lower variance than any point on the line. Temperature is a path through this space, not noise added at a fixed point.

3 Harmonic Latent Space

3.1 Entropy patching and the corpus beat

This run reads raw bytes through a byte-latent encoder [20]: the stream is segmented into patches at points of high next-byte entropy and compressed through that bottleneck, so the model spends capacity where the bytes are least predictable and coasts where they are not. There is no reconstructing codec here and no per-patch latent to phase-lock - but the same geometry appears one level up. The patch boundaries are content-adaptive, so the patch stream already carries the corpus's own rhythm: its cadence is the byte-entropy profile of the data. The harmonic head then writes its standing wave over those patch features, and the bias/variance reading is unchanged - a static spectrum is the average patch rhythm, an input-conditional modulation is the structured deviation. Where CALM phase-locks a latent to the beat, the byte-latent encoder lets the beat fall out of where it chose to cut.

4 The Scaling Conjecture

4.1 The watch

Hold the title literally for a moment. A watch is a mechanism of intricate, meshed gears, and what makes it a watch is that the gears turn *in unison*. Drive one and you drive them all, because the teeth couple them into a single body. Now try to run the watch by turning each gear in sequence - one at a time, the rest held still. You cannot. The logic does not merely become inefficient; it stops being a watch. The architecture, the mesh itself, forbids sequential operation. The coupling is the whole point.

Autoregressive consensus modeling is the attempt to turn the gears in sequence. It emits one position, conditions on it to emit the next, and accumulates the answer a tooth at a time. For a coupled system this is the wrong shape of computation, and its cost is the cost of enumeration: linear in the parts.

The harmonic field turns the gears in unison. The frozen Weyl phases are the teeth - the coupling, cut once and never re-cut, that ties every feature to every position in fixed relation. The amplitude grid is the single mainspring. One inverse transform releases it, and the entire field, every (t, d) cell at once, assumes its shape in a single coupled motion rather than a loop over positions. That is the mechanical content of "logarithmic": the whole mechanism moves together, so the cost tracks the number of independent rhythms it carries, not the number of positions it must cover.

And the field is *rebuilt*, not advanced. With input-conditional amplitudes it is re-derived from the input rather than carried forward and nudged; in the limit, the entire shape is reconstructed at every decision boundary, all of them, the whole computation re-formed around the present input. What the model learns under the smoothness prior - which asks the field at one position to predict the field at the next - is not a sequence of outputs but *the ability to hold a shape over time*: to keep the gears meshed, to keep the watch a watch, while the hands sweep. The frozen phases protect the coupling; the smoothness prior protects the form.

This is why the geometric framing is not ornament. Nearly anything we can imagine can be written as a shape - a configuration in some space, a standing wave, a mesh of constraints that must move together or not at all. A model that builds shapes and protects them over time models that structure directly, in unison, where a model that enumerates is forever turning one gear and hoping the rest follow. There is something here, and the rest of this section is an attempt to say how much.

The watch has a second axis of motion, and it is the one the architecture actually runs on. In recurrent mode Praxis does not stack a distinct gear per layer; it applies the same small mesh again and again - in calm-d, two physical layers turned over 4 recurrent steps - so depth is not a taller machine but the same machine revisited. ArcAttention and ArcGLU make the turning explicit: each carries a `current_depth` signal and warps its rotary phase and its gate per step, so the same weights present a different face at every revolution while the frozen coupling underneath is never re-cut.

This is the watch again, rotated into the depth dimension. At each step you turn the gear and force the geometric structure to comply: the field is not advanced but re-formed, re-asserted under a per-depth modulation rather than enumerated position by position. What moves is not the data through a pipeline but the shape itself, conforming at every turn to the focal point of the computation - the attention that constructs it. The recurrent step is the hand sweeping the dial; the mesh stays a mesh because the same teeth, revisited, hold it in unison across all 4 turns.

Read continuously, the dial is time itself: not a line of ticks but a full turn projected outward from the focal point over an interval, with the 4 recurrent steps as stages sampled around it. This is where XOR enters. XOR is the smallest function no linear model can separate, and its four configurations sit at the corners of a square - but lay those corners on a circle and parity becomes separable by a single phase, read off as position around the turn. The recurrent loop is an approximation of exactly that circle: four corners, sampled here in 4 turns of the same small mesh, each step advancing the angle toward the configuration the field must resolve. The model never computes XOR at a point; it rotates the whole field until the answer falls at the right phase - the gears turning, in depth, around a parity the geometry already knows how to hold.

Lemma 1. *For $a, b \in \{0, 1\}$ let $s = a + b$. Then* $\mathrm{XOR}(a, b) = \frac{1}{2}\big(1 - \cos \pi s\big)$*, so a single sinusoidal feature of s separates XOR - whereas no linear function of (a, b) can, XOR being the canonical non-separable case.*

Proof. $s \in \{0, 1, 2\}$ and $\cos \pi s = 1, -1, 1$ respectively, so $\frac{1}{2}(1 - \cos \pi s) = 0, 1, 0$ - exactly XOR on $(0, 0)$, on $(0, 1)$ and $(1, 0)$, and on $(1, 1)$. Placing the four configurations at angles πs on a circle thus makes parity a threshold on one phase coordinate. Linear inseparability of XOR in (a, b) is classical. $\square$

This is also why the bet is made small. Praxis is built and tested on the order of a million parameters - far below the scale where a model can simply store the corpus - because the claim is only meaningful where there are not enough parameters to memorize. At that size capacity cannot come from count, so it has to come from somewhere else, and the architecture's wager is that it comes from computation: the recurrent depth that turns the same small mesh again and again, and the harmonic coupling that moves every feature in one motion. Capacity is paid in the turning of the gears, not in their number.

The forced structure is the mechanism by which that payment buys anything. A harmonic field, a continuous latent, an autoregressive factorization - each is an inductive bias that makes the model compute the way a particular mathematics computes, importing that domain's structure rather than treating the architecture as neutral ground. Some of what is imported is exact: an autoregressive factorization is a proper normalized likelihood by the chain rule, not an approximation of one. Most of it is structure, not a theorem - a smoothness, a basis, a fixed point to iterate toward - and we keep the distinction honest. The stronger reading, that computing in a domain's mathematics confers that domain's guarantees, is the question the framework exists to test, not a result it assumes; the conjecture above is its measurable edge. If interference scales as $\log N$, a model with few parameters but many coupled turns should reach what a far larger consensus model reaches - which is exactly the experiment a million-parameter run is for.

The gear does not have to finish its turn. Praxis can halt the recurrent loop early, and the way it decides is the part most readers miss. After each turn the model compares the field to its previous state and asks one non-differentiable question - has the shape stopped changing? - by measuring the KL divergence between successive hidden states and stopping when that divergence collapses below a threshold. It is a hard, gradientless gate, decided under `no_grad` and never trained by descent: a discrete decision riding on top of a continuous field.

Read against the circle, that decision is ternary. One turn is unconditional - the loop always runs at least once, an always-true root that anchors the rest, the way XOR needs an always-on bias unit before two inputs can be separated at all. The remaining turns, all but the last of the 4 (the final boundary is forced), are the decisions the gate actually makes: two-valued comparisons over the interior of the circle that together approximate a parity - keep turning, or the shape is already held. And the test is inverted from the obvious one. The model does not confirm that an answer is right; it confirms that another turn would change nothing, validating by establishing that further computation would be wrong in exactly the expected way. Convergence is proven by non-change.

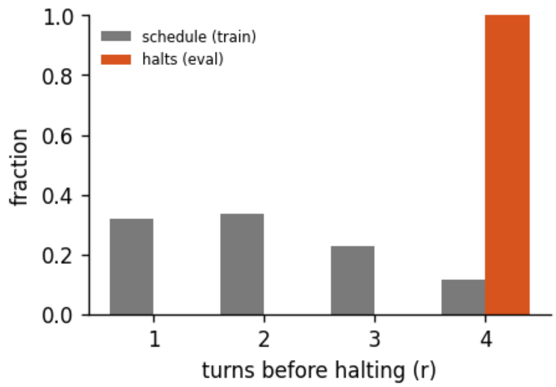

Figure 2: Halting distribution: the fraction of forward passes that stop after r turns, for the 4-turn recurrent loop of run calm-e, early-halt rate 0%, mean boundary KL 45.7. *schedule* is the random loop-count the model trains under; *halts* is where the KL gate actually stops at inference. A mass pinned at the maximum (KL never collapsing) is the honest signature of a model that has not yet learned to converge early - the decision is real, not yet decisive.

Why three, and not two? The reason is geometric. Two points fix only a line; three non-collinear points fix a plane (Lemma 3), and a plane is the least structure on which a phase can turn - a circle, a rotation, the standing wave the harmonic field is built from all live in two dimensions or not at all. A binary distinction names one of two points on a line: it can carry sign but never orientation. The third state lifts the representation off the line into the plane, where phase, and therefore harmony, first become possible. So the ternary gate is not a coarser version of a binary one; it is the minimal resolution at which the geometry can hold an oriented shape and let it harmonize - and here that resolution is read not off a hard bit but out of the continuous, lagged latent space the experts vote over.

This is why the halting signature is diagnostic rather than cosmetic. A model that has learned the geometry halts early, its mass sliding toward the low turns; one that has not keeps turning to the maximum, the gate firing at every boundary but never tripping. The pause is the model lifting itself out of the computation mid-turn, checking whether the shape it holds is the shape it would hold a turn later, and re-inserting itself only when the answer is no.

Lemma 2. *Over a recurrent loop of depth D that reuses L physical layers, the KL halting gate evaluates its divergence test at every loop boundary except the last, making exactly $D/L - 1$ binary decisions; the first turn is therefore unconditional.*

Proof. The decoder issues D expert calls and the gate fires only at a boundary, where $(\text{depth} + 1) \bmod L = 0$, of which there are D/L. At the final boundary the budget is exhausted and the gate returns without testing, leaving $D/L - 1$ divergence tests. The first turn precedes the first boundary, so it always runs. Hence one unconditional turn and $D/L - 1$ gated decisions. $\square$

4.2 Linear consensus, logarithmic interference

Here is the claim the rest of the paper has been building toward, stated as a conjecture because that is what it is.

A model that represents its corpus as a *mean* - a consensus, an average of examples - must spend resources roughly in proportion to the number of distinct patterns it needs to express. To hold N patterns it allocates capacity that grows with N. This is the linear regime, and it is consistent with why scaling laws ask for ever more parameters and tokens to buy each further increment of capability [15].

A model that represents its corpus as *interference* - a superposition of K frequency components - addresses a combinatorially larger space. K components with learnable amplitudes and frozen, equidistributed phases produce a field whose distinguishable configurations grow exponentially in K, because it is the *pattern of constructive and destructive interference* across components that carries information, not the components individually. To express N configurations you need on the order of $\log N$ components. This is the logarithmic regime. The same asymmetry shows up in time: the transform that builds the field is $O(n \log n)$, and a single transform realizes the whole interference pattern at once rather than enumerating its parts.

4.3 The exponential edge

Why should a learned network be able to live in the interference regime at all? Because the transformer already computes on an exponential substrate. Every attention layer is a softmax, $\exp(\cdot)/\sum \exp(\cdot)$, which amplifies small differences and converts addition into multiplication; the network is kept numerically alive only by normalization holding activations in the narrow band where the exponential is both sensitive and stable. Euler's identity, $e^{i\pi} = -1$, ties that exponential directly to periodic structure, which is why a harmonic field seeded with π and e is not an arbitrary choice of decoration but a resonance with the machinery already present. The harmonic head writes interference patterns onto a substrate that was always exponential; it makes explicit a structure the softmax was implicitly using.

4.4 Tractability is observer-relative

If interference lets a model represent exponentially many configurations with logarithmically many components, then a search that is combinatorial when enumerated becomes, in this representation, the reading-off of an interference pattern computed in a single transform. The hardness did not vanish. It moved

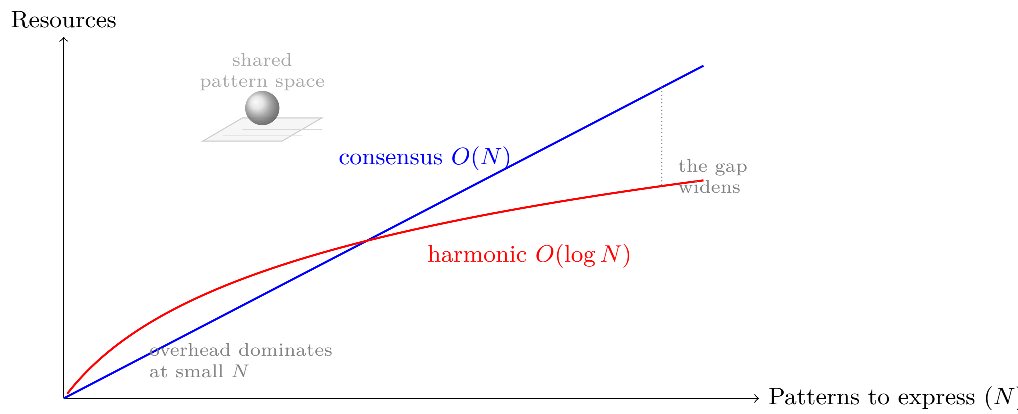

Figure 3: Conjectured scaling regimes. To express N patterns, consensus modeling - learning the mean - spends resources growing linearly in N (blue). Harmonic modeling composes patterns by interference, and $K \sim \log N$ components suffice, so its cost grows logarithmically (red). The harmonic basis carries fixed overhead, so it loses at small N and wins at scale, where the gap widens without bound. Both must ultimately represent the same target (the shared pattern space, top); the claim is about the cost of getting there, not the destination. This figure states a conjecture, and replaces the speculative measured-curve figure of earlier drafts.

into the choice of basis. And that relocation is the real claim, which we will state without hedging: in a continuous, nested representational space, P and NP are not two discrete bins a problem falls into once and for all. They are the ends of a spectrum, and where a given instance sits on that spectrum is set by the representation doing the solving - by the observer.

This is the same thesis the rest of the paper has been making, carried to its conclusion. If attention constructs which patterns manifest (Section 6), and if the basis a model occupies decides whether a structure costs linear or logarithmic resources to express (Figure 3), then "intractable" cannot be a property of the problem alone. It is a property of the pairing between a problem and a solver. The instance that is exponential to one architecture - one that can only enumerate, that holds no basis in which the instance is smooth - is approximable in near-constant work by another that happens to carry the right harmonic. Some observers are simply better at collapsing NP-shaped problems toward P; others are blocked, not by the problem, but by their own architecture, perspective, or alignment. The watchmaker who cannot perceive the gear cannot build the watch, however long he is given. The difficulty was never only in the watch.

We keep exactly one boundary, and we keep it because it sharpens the claim rather than softening it. This is not a statement about the formal worst-case question - whether the classes P and NP coincide over all inputs and all machines [6]. That question is deliberately indifferent to who is asking; it abstracts the observer away on purpose. The claim here is the complementary one, and abstracting the observer away is precisely what it refuses to do: for the bounded, structured, approximate problems that real minds and real models actually face, tractability is observer-relative, continuous, and movable - and moving it is what changing your representation *is*. The weak, measurable form is already in the framework: for a family of problems one can record the scaling slope each architecture buys, and watch a single instance migrate along the P-to-NP spectrum as the basis changes underneath it.

5 The Praxis Framework

Praxis [4] is a registry-based framework in which every component - encoder, decoder, attention, optimizer, loss, router, embedding, head, memory - is swappable, and new implementations become available through registration without touching core infrastructure. Read in light of the preceding sections, its components are not an arbitrary catalogue; each one shapes the representational geometry in a different way, and the framework's purpose is to let those geometries be combined and compared rather than to scale any one of them.

- **ByteLatent compression** [20] forces re-attention through an entropy-based patching bottleneck, selecting which correlations survive aggressive dimensionality reduction.
- **MonoForward training** [18] detaches gradients between layers and trains each with a local objective, turning one global trajectory into many independent ones - whether run in-process, across a Ray actor per layer, or over the remote-expert pool, where a slow peer never blocks the rest.
- **Reversible residuals** [11] require invertibility throughout the forward pass, forbidding destructive compression and so forcing reconstruction-preserving representations.
- **Attention variants** (multi-head, sliding-window, gated, structured $O(n \cdot c)$, and others) each impose a different relational geometry on the sequence.
- **Mixture-of-experts routing** [23] and **tool integration** let several geometries coexist and let external, bias-free computation enter the loop.

The configuration hierarchy (Environments > CLI > Experiments > Defaults) makes arbitrary combinations reproducible: different attention per expert, optimizers per layer, frequency offsets [17, 27] per head. 28 training runs are documented in the run history. This paper does not present a benchmark table, by design: the framework is the instrument, and the experiment that matters is the one the reader runs to test the scaling conjecture for themselves.

6 Attention as a Constructor

The framing that motivates all of this is older than the harmonic head. Across physics and neuroscience, observation appears to participate in what is observed rather than passively recording it: measurement affects the quantum system [13], attention modulates cortical responses to identical stimuli [29, 16], and predictive-processing accounts make perception an attention-weighted construction rather than a bottom-up readout [10, 5]. If attention constructs which patterns manifest rather than filtering pre-existing ones, then architectural diversity in attention is not a tuning detail; it determines what is learnable at all.

The measurement analogy invites a sharper question than the one it usually settles for: if observation participates in the observed, does *our* act of observing the model - snapshotting its spectrum, sampling its state - change what it learns? The quantum reading is metaphor, and we keep it as such. But a harmonic field is a standing wave over the full (position, feature) computation, and therefore non-local: a perturbation injected at one cell rides the field to the rest rather than staying put. So a harmonic model should be *more* sensitive to impure observation than one whose features are independent, and the weak, measurable form of the question follows directly. Two of its artifacts are mundane and real. First, a measurement that mutates state rather than reading it purely is a literal observer effect - we have already seen one, where a background thread flipped the model's train/eval flag mid-forward and changed gradients that gate on it; harmonics propagate such a perturbation instead of localizing it. Second, observation forces dtype boundaries a pure forward pass avoids (snapshots cast; the inverse transform is sensitive in its low-order bits), and on the exponential edge of Section 4 rounding can amplify rather than wash out - precisely where the field is most plastic. Both are testable as a diff between a watched run and an unwatched one from the same seed, against a non-harmonic control that isolates the field's non-locality. The strong form is the architecture's own: the remote-expert pool the framework runs is, by construction, a passive observer of the model's batches whose gradients never return, so drawing that single coupling - observer to participant - is a clean two-arm ablation of whether being watched, when watching also feeds back, changes which patterns manifest.

7 Discussion

7.1 Answering the opening question

Do you exist in other people's realities the same way you exist in your own? If the geometry of an attention architecture determines which patterns it constructs, then observers running different geometries build different representations from the same input. You exist in your reality through your geometry, and in another's through theirs. This is not solipsism: convergence to a shared pattern space (Figure 3) says everyone is reaching the same place. It is the *path* that differs - which patterns manifest, in what order, at what cost - and the path is set by architecture.

7.2 A note on reconciliation

The same convergence principle drives the author's companion infrastructure project, Platformer [3], where the explicit design stance is eventual consistency: rather than seeking perfect synchronization, the system runs repeated reconciliation cycles that drift toward a declared desired state and "find balance" through iteration. The shift there from fragmented to unified state is described in the same terms as the shift from sparse to fully-connected networks - more global context yields better optimization. Training is reconciliation of this kind: a model is run against its objective again and again until it settles into an attractor, the way a harmonic field settles onto a corpus rhythm or a codec phase-locks. That the same philosophy organizes both a language model and a fleet of cloud accounts is not a coincidence of the author's taste; it is what convergence under repeated local correction looks like, in whatever substrate.

7.3 The practitioner as proof

The blind watchmaker is design without a designer: structure that emerges from a blind, iterative process rather than from foresight. That is what these systems do. No one specifies the spectrum the harmonic field will lock onto, or the geometry the crystal centers will grow into; they emerge from the process, and we read them off afterward. Which raises an honest question about what would even count as proof here. A benchmark number proves that one model beat another on one distribution. It does not establish a claim about regimes of representation. The framework's proof, if it has one, is of a different kind: it is the practice itself. *Praxis* is the Greek word for theory enacted - for doing as a form of knowing. The demonstration on offer is not a leaderboard but a method you can run, whose intuitions arrive only by stepping through it yourself. The name is the argument: the proof is in the doing.

7.4 Implications for machine learning

If bias and variance are separable in a harmonic latent space, then they should be learned as input-dependent functions rather than chosen as a global operating point: the field learns which rhythm matters, the delta learns when and how much to deviate, and no per-experiment hyperparameter mediates the two. If the scaling conjecture holds even partially, then on problems with strong periodic or compositional structure, the right move is to change the basis rather than to add parameters, and the linear and logarithmic regimes should be distinguishable by measuring the slope of coverage against resources. Both are testable in the framework, which is the only reason to state them.

7.5 Why architecture is not neutral ground

A standing objection deserves a direct answer, because the framework's whole premise stands or falls on it. If universal approximation [14] guarantees that a sufficiently wide network of *any* family can represent any function we care about, then any Praxis experiment - however intricate - is by hypothesis emulable by some other network, and one is entitled to ask where the value of the architectural complexity could possibly live. The answer is that the objection conflates three distinct equivalences, and universal approximation establishes only the first of them.

Representational equivalence says the function exists in the closure of each family; it is an existence statement at unbounded width. Learnability equivalence would say stochastic gradient descent, from a random initialization, actually *reaches* that function in feasible compute and data. Statistical equivalence would say two families generalize equally at a fixed sample budget. Universal approximation asserts the first and is silent on the other two by construction - it quantifies existentially over parameters, not over what an optimizer finds or what a finite sample supports. The value of an architecture lives entirely in the second and third layers, which is exactly where the existence theorem says nothing.

The foundational mathematics here cuts *for* the framework, not against it. Depth-separation results (Proposition 1) exhibit functions whose representation cost differs exponentially between architectures that are equal as sets of representable functions: the family is the same in the limit, the resource at fixed accuracy is separated by $2^{\Omega(k)}$. "Architecture is just a constant factor" is therefore not a conservative default but a claim already contradicted by theory; the burden of proof runs the other way. The honest residue is that representation cost is not yet learnability, and proving that a given pair of comparable-capacity architectures have *separated error boundaries under training* - the empirical edge of the conjecture in Figure 3 - requires controlled ablations we have not yet run. We state the hypothesis before we have closed it, and mark the seam plainly: the separation of representable cost is established mathematics; the separation of *learned* error across configurations of equal budget is the measurable prediction the framework exists to test, not a result it already owns.

Proposition 1. *Let $\mathcal{F}_A$ denote the set of functions a family of networks with architecture A can approximate to arbitrary accuracy given unbounded width. Universal approximation [14] asserts $\mathcal{F}_A = \mathcal{F}_B$ for any two such families: as* sets of representable functions *they are equal. This equality does not imply equality of* cost *- the width, depth, or sample budget at which a target is reached - and there exist target functions for which the cost gap between two architectures is exponential.*

Proof. The first claim is immediate: universal approximation quantifies existentially over parameters at unbounded width, so it constrains only which functions lie in the closure of each family, not the resources at which they are reached. Equality of sets is compatible with any cost profile whatsoever.

The separation is witnessed, not asserted. Telgarsky [28] exhibits functions on $[0, 1]$ that a network of depth $\Theta(k^2)$ and constant width represents exactly, but that any network of depth $O(k)$ requires $2^{\Omega(k)}$ width to approximate. Eldan and Shamir [8] exhibit a function expressible by a depth-three network of polynomial width that any depth-two network must use exponential width to approximate. In both, $\mathcal{F}_A = \mathcal{F}_B$ holds in the unbounded limit while the resource at fixed accuracy is separated exponentially. Hence representational equivalence is the wrong invariant for comparing architectures of bounded size; the relevant quantity is cost, and cost is architecture- dependent by these results. The further descent - from representation cost to what stochastic gradient descent actually *reaches* from random initialization, and to what *generalizes* at a fixed sample budget - is the inductive-bias question, on which approximation theory is silent by construction; the lottery-ticket phenomenon [9] is one empirical marker that two equal-capacity families do not induce the same reachable solutions. We import these as the foundation the framework's conjecture stands on, not as results re-derived here. □

7.6 Neither halting nor blocking

A recurring objection holds that a neural network cannot be Turing-complete, and the halting problem is offered as the reason. The invocation is a category error. Turing-completeness is the capacity to express any computation, including ones that never halt; deciding in advance whether an arbitrary program halts is a separate question, and an undecidable one for Turing machines themselves. The theory in fact runs the other way - recurrent networks with unbounded precision and unbounded time are already Turing-complete [24]. What bounds a real model is not undecidability but finite precision and a finite number of steps, exactly the resources a fixed-depth forward pass exhausts and exactly the ones recurrence and delay restore.

Praxis leans into that. With the swarm online, computation runs in two directions at once. Forward, the recurrent loop propels generation and is never required to emit a stop: the halting gate is a convergence check, not an exit, so generation is a continuous autonomous stream rather than a program that terminates. Sideways, a pool of remote experts votes - stochastically sampled, mixed by a CALM-style expert vote - and trains without blocking, so a slow peer never stalls the others; its update simply lags and lands on a later pass. Computation is lagged as much as propelled, and the model commits to neither a halt nor a barrier. It approximates both by doing neither.

This relocates the question. It is no longer whether the network halts but what the pool expresses, over how much time, at what precision - the resources, not the stop bit. The butterfly is the right image: one expert, then two, then four, each a wing whose small vote the mixture amplifies or cancels, the whole field sensitive to where the late votes fall. The interior turns of the recurrent loop (Section 4) already carry the decision; the last is only the budget running out. Power here is bought with peers and patience, not with a proof that the machine knows when to stop.

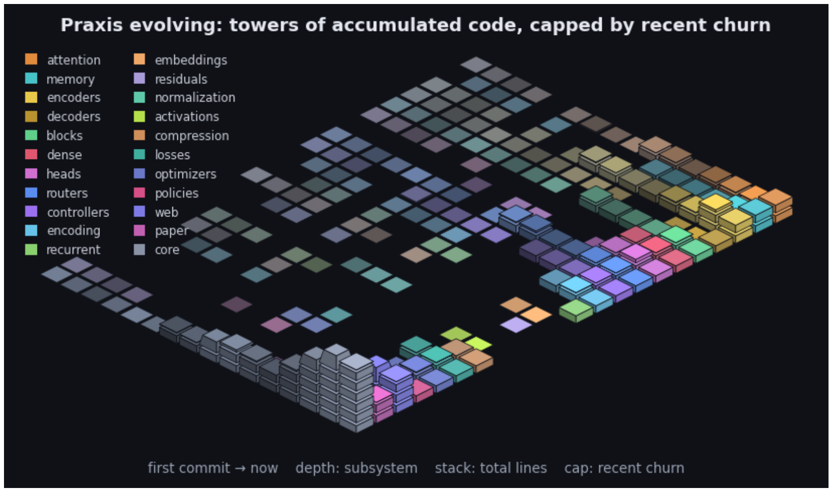

Figure 4: Praxis's git history as an isometric terrain over the last 2245 commits. Each cell is one subsystem in one time window: a stack of base blocks for its accumulated size (total lines, recency-weighted) topped by a vivid cap for that window's line churn - the standing codebase carrying the recent activity. Both are weighted by recency (the kernel the model applies to a sequence, turned on the repository): recent windows stack tall with bright caps while history thins toward the always-present prior - the low valley we roll into. Color fades from each subsystem's hue toward a neutral prior into the past, banded in phased strata over a non-linear timescale. It is the bias/variance geometry read along the frequency axis (see the project notes): loud recent corrections over the quiet, ever-present base. By recency-weighted churn the current center of gravity is core, web, encoders. The framework charts the hand that builds it.

A living document can also look at itself. Praxis is built in the open, commit by commit, and that history is just another sequence - a stream of byte-level diffs over time. Figure 4 turns the framework's own recency kernel on its repository: per-subsystem churn across the project's history, with the distant past faded in proportion to its distance from the present, the way a decay bias weights earlier positions in a sequence. It is not an argument, only a mirror - the same machinery that reads a corpus, reading the record of its own construction. The reading that any finite byte sequence is a position to be classified the same way, the repository included, is left as a conjecture to test rather than a claim to lean on (see the project notes).

7.7 Standing conjectures

We gather the paper's falsifiable conjectures in one place; each is discussed where it arises. They are bets about reachability and cost, not theorems, and each names what would refute it.

- **Linear consensus, logarithmic interference.** Learning a corpus as a consensus mean costs resources linearly in the patterns expressed, while composing it by harmonic interference costs only logarithmically, so the coverage-versus-resources slope a basis buys is measurable and basis-dependent. [15]
- **Reachable, not merely representable.** Across architectures of comparable parameter count and compute there exist task and data regimes where generalization error is separated, because the prior sets which solutions stochastic gradient descent can reach, not only which functions exist in the closure. [28, 9]
- **Lottery engineering.** A geometric prior makes good sparse subnetworks reachable by stochastic gradient descent directly rather than only by train-then-prune, and a rotating population of such subnetworks (mixture-of-widths) approaches full-width performance at lower average active width. [9]

7.8 Limitations

We are deliberate about what is and is not established. The bias-variance decoupling is grounded in a running implementation and its diagnostics (concentration, smoothness, delta norm, the spectrum heatmap), but "orthogonal axes" is an interpretation of those diagnostics, not a theorem. The scaling conjecture of Figure 3 is a conjecture: the figure draws an asserted relationship, not measured curves, and the honest next step is to measure the slope on tasks of varying combinatorial structure and report where harmonic representations fail to buy it. The observer-relative view of tractability (Section 4) is a philosophical position, and only its weak form - the scaling slope a given basis buys on a given problem family - is measurable; we make no claim about the formal worst-case P versus NP question, which abstracts the observer away by design. The connection to biological attention remains speculative, and the prismatic asymmetry needs to be shown in silico now that the hand-measured figure is gone. The observer effect of Section 6 is, in its quantum form, an analogy we do not cash; only its weak form is a claim - that impure measurement (state-mutating reads, measurement-induced rounding on the exponential edge) perturbs a non-local harmonic field measurably more than an independent-feature one, and that coupling the framework's passive observer pool back into the forward is a two-arm ablation of being-watched. Those are diffs between a watched and an unwatched run, not statements about physics. None of these are reasons not to state the ideas; they are the boundary between what we have built and what we are conjecturing, drawn so the reader can attack the right target.

8 Conclusion

We began with scale and ended with geometry. The patterns a model can learn are set by the shape of the space its weights may occupy, and a harmonic latent space reshapes the oldest tradeoff in the field: bias becomes a static, population-average spectrum, variance becomes an input-conditional modulation, and because they live on orthogonal axes they can be lowered together. From that decoupling follows a conjecture stated plainly - that learning the mean costs linearly while composing by interference costs logarithmically - and a framework built to test it rather than to argue it. And from that, in turn, follows the harder claim: that tractability is not a fixed property of a problem but a relation between a problem and the observer solving it. The same instance is exponential to one architecture and nearly free to another, so P and NP are better read as a spectrum that the choice of representation indexes than as a wall every solver meets in the same place. The speculative figure is gone; the conjectures that replace it are falsifiable. What remains is praxis: the answer manifests through direct experimentation, and what you choose to attend to determines what we find next.

One reading the geometry keeps inviting is worth stating as we close, marked as a direction rather than a result. The case for the third state was that two points fix only a line and three fix a plane (Lemma 3), and that a phase can turn only in a plane - so the least representation that can hold an oriented shape is two-dimensional, not one. But that plane is only where a phase *sits*. A phase also moves: it carries a velocity and a direction, and a position taken together with its motion is a phase space, twice the dimension of the configuration it lives on. The line becomes a plane, and the plane the harmonic field was built to turn in becomes the four-dimensional space in which that turning is itself a coordinate. The representation we have described is therefore never a point; it is a moving point, and the climb - from a position, to a line, to the plane where phase lives, to the phase space where phase moves - is the same ascent at every scale. We make no claim that the network parameterizes that space explicitly. We note only that the geometry, followed honestly, does not stop at the plane: velocity and direction, not position alone, are what a harmonic representation is finally about.

Lemma 3. *Two distinct points determine a line; three non-collinear points determine a unique plane. A phase - a point on the circle S^1 - is realizable only in dimension at least two, so three points are the minimal affine data that orient the plane a harmonic component turns in.*

Proof. Through p_0, p_1, p_2 the affine span $\{\, p_0 + s(p_1 - p_0) + t(p_2 - p_0) \,\}$ is a plane exactly when $p_1 - p_0$ and $p_2 - p_0$ are linearly independent - that is, when the three points are non-collinear; two points give a single direction and hence only a line. Since S^1 embeds in $\mathbb{R}^2$ but in no line, a continuous phase requires that plane, and three points are the fewest that fix it. $\square$

A Current Experiment

This appendix is generated. The figure is built from training logs by `praxis/pillars/build.py`; the paper itself is a template the tool populates. The chart leads with the most recently launched experiment that has logged validation data, and fixes the y-metric to that run's family - here Validation Loss (`val_loss`), as of the last export (2026-06-06). Only experiments that report the *same* metric are drawn alongside it; the rest are listed as omitted below. This is the smallest living slice of the document: one metric, the current experiment against its compatible predecessors.

Note: The current experiment (calm-e) has not logged usable validation metrics yet; the chart leads with the most recent experiment that has (calm-d).

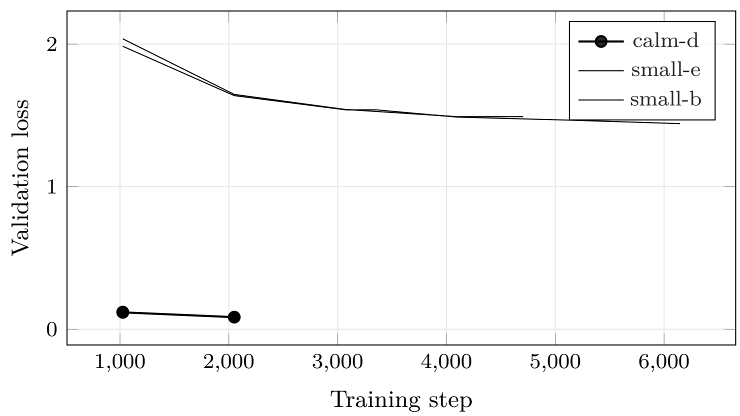

Figure 5: Validation Loss across the 3 most recent experiments with comparable validation curves; the leading run (calm-d) is drawn in bold. Generated from run logs - metrics that are not comparable across families (token loss vs. byte bits-per-byte vs. codec/CALM BrierLM) are excluded by construction, so the curves on this axis mean the same thing.

The current experiment, calm-d (`68f1be505`), reached Validation Loss 0.08464 by step 2050. Recent experiments whose families report a different validation metric are omitted rather than plotted on an incomparable axis: calm-e, calm-f, beta, calm-c, calm-b, calm-a, wavelet, small-d, small-c.

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